Uniform absorption asymmetry on a fixed frequency band

A constructive draft for a strict extinction floor

Research working note

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Abstract

We give a proposed constructive extension of the compact two-layer prescribed-extinction argument to a fixed frequency band. For every band bounded away from zero, every positive center-frequency extinction, and every strictly smaller positive extinction floor, the argument yields compact reciprocal nonmagnetic dielectric bodies whose worst-case normalized absorption asymmetry approaches one. Each body has a single passive causal constitutive law, made from finitely many Lorentz oscillators, and obeys arbitrary prescribed upper dielectric and loss caps throughout the band. The additional ingredient is a passive matrix-valued dispersion construction that shapes polarization conversion; a fixed-axis weak plate alone gives a weaker floor range. The argument is an analytical research draft. It is separate from the existing manuscript and frozen formalization. The boundary case in which the floor equals the prescribed center extinction is not established.

1 Statement and exact scope

Fix

$$B = [\omega_1, \omega_2], \qquad 0 < \omega_1 < \omega_c < \omega_2, \qquad k = \omega/c_{\mathrm{v}},$$

where c_{v} is the vacuum speed. Use the opposite incident electric fields $f_{\pm} = e_y \mathrm{e}^{\pm ikz}$ with equal incident intensity. The object comprises two homogeneous layers with common square aperture $S_R = (-R/2, R/2)^2$, thicknesses $h, d > 0$, and vacuum outside. Each layer has a spatially constant, symmetric 3×3 dielectric tensor depending on frequency; $\mu = I$. The total extinction, absorption and full-sphere scattering cross sections satisfy

$$X_+ = X_- = X = A_{\pm} + S_{\pm}, \qquad \Gamma = \frac{|A_+ - A_-|}{X}.$$

These are plane-wave cross sections, not fractions of a finite incident power.

Proposed theorem 1 (Fixed band with a strict extinction floor)**.** *Fix $P > 1$, $Q > 0$, $E_* > 0$, and $0 < E_{\min} < E_*$. For every $0 < \epsilon < 1$, the construction below supplies a finite two-layer body and one passive causal reciprocal dielectric law in each layer such that, for every $\omega \in B$,*

$$I \preceq \mathrm{Re}_H\, \varepsilon(\omega) \preceq PI, \qquad 0 \prec \mathrm{Im}_H\, \varepsilon(\omega) \preceq QI, \tag{1}$$

$$X(\omega_c) = E_*, \qquad X(\omega) \geq E_{\min}, \qquad 1 - \epsilon < \Gamma(\omega) < 1. \tag{2}$$

Each susceptibility is a finite sum of damped Lorentz oscillators with real positive semidefinite matrix residues, and tends to zero at infinite frequency. Consequently, within this resource-unrestricted class,

$$\sup_{\textit{admissible bodies}} \inf_{\omega \in B} \Gamma(\omega) = 1.$$

The supremum is not attained.

Material caps are imposed on B, not on all real frequencies. There is no common aperture, thickness, volume, oscillator-strength, number-of-oscillators or positive-loss-floor bound across the sequence. Frequencies do not select different bodies: all geometrical and oscillator parameters are fixed for each selected body. The constituent oscillator axes are fixed; the principal axes of their summed tensor may vary with frequency. This is permitted by the two-layer tensor class of the base manuscript [1].

The proof uses the weak-Maxwell volume representation, radiation identity and optical theorem established in the base manuscript. The new frequency-uniform localization, matrix factorization, dispersion shaping and parameter selection are supplied here. This is not a new Lean verification or an independent audit of all inherited Maxwell foundations.

2 Passive dispersion can shape the conversion amplitude

For $F > 0$, $\Omega > 0$, $\gamma > 0$, define

$$\ell(z) = \frac{F}{\Omega^2 - z^2 - i\gamma z}.$$

Its poles lie in the lower half-plane, it obeys the real-time symmetry $\ell(-\overline{z}) = \overline{\ell(z)}$, and for $\Im z > 0$,

$$\Im\{z\ell(z)\} = F\frac{(\Im z)(\Omega^2 + |z|^2) + \gamma\,|z|^2}{|\Omega^2 - z^2 - i\gamma z|^2} > 0. \tag{3}$$

Thus $z\ell(z)$ is Herglotz, the passivity condition for electric susceptibility. Multiplication by a real positive semidefinite matrix, finite summation, and congruence by a fixed real matrix preserve this condition. If $\Omega > \omega_2$, both real and imaginary parts of $\ell(\omega)$ are positive throughout B. Matrix oscillator positivity is also central in general broadband scattering representations [2]; the local construction used here is explicit.

Lemma 1 (Signed resolvent approximation)**.** *The real linear span of $(t-x)^{-1}$, $t > T > \omega_2^2$, is dense in $C([\omega_1^2, \omega_2^2]; \mathbb{R})$.*

Proof. An annihilating finite signed measure μ would have Stieltjes transform $\int (t-x)^{-1}\,\mathrm{d}\mu(x) = 0$ for every $t > T$. Expanding at infinity gives $\int x^n\,\mathrm{d}\mu(x) = 0$ for every nonnegative integer n. Polynomial density on the compact interval implies $\mu = 0$. The Hahn–Banach theorem and the Riesz representation theorem now give density. Every approximant is a finite sum; repeated poles may be merged. □

Fix $a_0 > 0$ and choose

$$\tau = \frac{2\pi}{\omega_1 + \omega_2}. \tag{4}$$

In particular $0 < \tau\omega_1 < \tau\omega_2 < 2\pi$. The positive continuous target

$$f(\omega) = \frac{a_0}{2\sin(\tau\omega/2)}$$

is bounded on B. Lemma 1 gives a finite real rational sum

$$r(\omega) = \sum_{j=1}^{m} \frac{\beta_j}{t_j - \omega^2}$$

arbitrarily close to f. The β_j may have either sign. For any $\eta_j > 0$, the residue

$$M_j = \begin{pmatrix} \eta_j & \beta_j \\ \beta_j & \beta_j^2/\eta_j \end{pmatrix}$$

is positive semidefinite. Choose the η_j small enough that $\sum_j \eta_j/(t_j - \omega_2^2)$ is arbitrarily small. Define the lossless intermediate matrix

$$C_0(\omega) = \frac{a_0 T_0}{T_0 - \omega^2} e_x e_x^T + \sum_{j=1}^{m} \frac{M_j}{t_j - \omega^2} + \frac{\kappa I_2}{T_* - \omega^2} = \begin{pmatrix} a(\omega) & b(\omega) \\ b(\omega) & c(\omega) \end{pmatrix}. \tag{5}$$

All poles exceed ω_2^2. Take T_0 sufficiently large and $\kappa > 0$ sufficiently small. Then $C_0 \succ 0$, a is arbitrarily close to a_0, and $b = r$ is arbitrarily close to f. The entry c may be extremely large, but it is finite and continuous. Its unrestricted size is an essential resource in this construction.

Replace every denominator in (5) by $t - \omega^2 - i\gamma_t\omega$, using positive damping also for the isotropic term. The resulting C_γ obeys

$$\mathrm{Re}_H\, C_\gamma \succ 0, \qquad \mathrm{Im}_H\, C_\gamma \succ 0, \qquad C_\gamma \longrightarrow C_0$$

uniformly on B as all dampings tend to zero. It is a finite passive reciprocal oscillator sum with high-frequency decay. All these choices describe one material law across the whole band.

3 The weak thick plate and the selective absorber

Set $D_\theta = \mathrm{diag}(1, \theta)$, $0 \le \theta \le 1$. Before final isotropic regularization, use the plate susceptibility

$$\chi_{W,N,\theta}(\omega) = \frac{1}{N} D_\theta C_\gamma(\omega) D_\theta \quad \text{in the } xy \text{ block}, \qquad d_N = \frac{2c_{\mathrm{v}}\tau N}{a_0}, \tag{6}$$

with zero z susceptibility. The absorber has susceptibility

$$\chi_{A,N}(z) = \alpha_{A,N}(z) e_x e_x^T, \quad \alpha_{A,N}(z) = \frac{4\omega_2^2/N}{4\omega_2^2 - z^2 - i\omega_2 z}, \quad h_N = \frac{N^2 c_{\mathrm{v}}}{\omega_c}. \tag{7}$$

For each fixed oscillator matrix, sufficiently large N gives strict upper margins in both material caps, uniformly in $\theta \in [0, 1]$ and $\omega \in B$. The lower real bound follows from positive real oscillator coefficients and positive semidefinite residues.

3.1 Uniform absorber limit

Write $x = \omega/\omega_2$ and $a_B = \omega_1/\omega_2 > 0$. Then $\alpha_{A,N} = 4/[N(4 - x^2 - ix)]$. On B,

$$|\alpha_{A,N}| \le \frac{4}{3N}, \qquad \Im\alpha_{A,N} \ge \frac{a_B}{4N}.$$

For the passive square root $n_A = \sqrt{1 + \alpha_{A,N}}$, $\Im n_A \ge c_A/N$ with $c_A > 0$ independent of ω. The slab propagation amplitude $p_A = \exp(ikn_A h_N)$ therefore satisfies $|p_A| \le \exp(-cN)$. Its Fresnel amplitude is $O(N^{-1})$. The exact scalar slab formulas give, uniformly on B,

$$r_A = O(N^{-1}), \qquad t_A = O(\exp(-cN)). \tag{8}$$

The y channel remains vacuum-equivalent in this intermediate absorber.

3.2 Uniform plate limit after normalization

First take $C_\gamma = C_0$, and put $M_\theta = D_\theta C_0 D_\theta$ and $t = \tau\omega/a_0$. The index matrix is $n_N = (I + M_\theta/N)^{1/2}$. All its slab amplitudes are matrix functions of M_θ:

$$\begin{aligned} r_0 &= (I - n_N)(I + n_N)^{-1}, & p_N &= \exp(ikd_N n_N), \\ R_W &= r_0(I - p_N^2)(I - r_0^2 p_N^2)^{-1}, & T_W &= (I - r_0^2)p_N(I - r_0^2 p_N^2)^{-1}. \end{aligned} \tag{9}$$

For fixed C_0, these expressions are holomorphic in θ on a fixed complex disk for all sufficiently large N. There,

$$n_N = I + \frac{M_\theta}{2N} + O(N^{-2}), \quad R_W = O(N^{-1}), \quad \frac{T_W}{v_W} = Q_\theta + O(N^{-1}), \quad Q_\theta = \exp(itM_\theta), \quad v_W = \mathrm{e}^{ikd_N}.$$

The estimates are uniform in real frequency on B and on the complex θ disk. Although complex θ is only an analytic device, its propagation matrices have a finite uniform norm on the disk; hence the denominators in (9) are invertible for large N.

Parity under $\theta \mapsto -\theta$ makes offdiagonal entries odd and diagonal entries even. At $\theta = 0$ the yy slab is exactly vacuum. Applying Cauchy estimates on a smaller disk therefore gives

$$\begin{aligned} (R_W)_{xy}/\theta &= O(N^{-1}), & (R_W)_{yy}/\theta^2 &= O(N^{-1}), \\ (T_W/v_W - Q_\theta)_{xy}/\theta &= O(N^{-1}), & (T_W/v_W - Q_\theta)_{yy}/\theta^2 &= O(N^{-1}). \end{aligned} \tag{10}$$

These are the needed normalized estimates; unnormalized convergence alone would not control vanishing coupling.

Direct expansion of the exponential, with $\phi = ta$, gives

$$(Q_\theta)_{xy} = \theta\frac{b}{a}(\mathrm{e}^{i\phi} - 1) + O(\theta^3), \tag{11}$$

$$(Q_\theta)_{yy} = 1 + \theta^2\left[itc + \frac{b^2}{a^2}(\mathrm{e}^{i\phi} - 1 - i\phi)\right] + O(\theta^4). \tag{12}$$

The large c contributes only an imaginary quadratic forward phase and does not enter the quadratic extinction coefficient.

3.3 Actual stack amplitudes and the normalized spectrum

Let $P_x = e_x e_x^T$, $P_y = e_y e_y^T$, $v_A = \mathrm{e}^{ikh_N}$, $R_A = r_A P_x$, $T_A = t_A P_x + v_A P_y$. Exact cascade matching gives

$$\begin{aligned} t_+ &= T_W(I - R_A R_W)^{-1} T_A e_y, \\ r_+ &= T_A R_W (I - R_A R_W)^{-1} T_A e_y, \\ t_- &= T_A(I - R_W R_A)^{-1} T_W e_y, \\ r_- &= \left[R_W + T_W R_A (I - R_W R_A)^{-1} T_W\right] e_y. \end{aligned}$$

The unlisted direct $R_A e_y$ term is zero. Reciprocal transmission gives the same co-polar forward coefficient. With $v = v_A v_W$, the planar extinction is

$$X_\infty = 2\Re(1 - e_y^T t_\pm / v),$$

and scattered transmission is $t_\pm - ve_y$. Define $Y_\infty = X_\infty/\theta^2$, and similarly normalized scattering $C_\infty^\pm$ and absorption $B_\infty^\pm$, with $\gamma_\infty = \left|B_\infty^+ - B_\infty^-\right|/Y_\infty$. Equations (8) and (10) justify their limits at $\theta = 0$.

In the ideal selective-absorber and reflectionless-plate limit,

$$t_+ = vQ_\theta e_y, \qquad t_- = vP_yQ_\theta e_y, \qquad r_\pm = 0.$$

Equations (11)–(12), or unitarity of Q_θ, give

$$Y_0(\omega) = 4\left(\frac{b(\omega)}{a(\omega)}\right)^2 \sin^2\!\left(\frac{\tau\omega a(\omega)}{2a_0}\right), \quad C_0^+ = Y_0, \quad C_0^- = 0. \tag{13}$$

Thus $B_0^+ = 0$, $B_0^- = Y_0$ and the limiting contrast is one. This is an actual dielectric-stack calculation, not a choice of independent scattering matrices.

The choices $a \approx a_0$ and $b \approx a_0/[2\sin(\tau\omega/2)]$ make Y_0 arbitrarily close to one uniformly on B. Positivity of the sine on the compact band makes this approximation stable. For each fixed lossless matrix, first choose finite N sufficiently large. Next choose positive damping sufficiently small. The exact finite-layer amplitudes and their normalized Taylor coefficients depend continuously on that damping. Real-frequency outgoing uniqueness follows from the passive flux identity and finite-interval ODE uniqueness, so no real pole obstructs the perturbation. We obtain a fixed, strictly damped active planar stage whose $Y_\infty(\omega, 0)$ is arbitrarily close to one and whose $\gamma_\infty(\omega, 0)$ is arbitrarily close to one, uniformly on B.

4 Matrix factorization at vanishing coupling

For the compact aperture, let

$$\mathcal{H}_R = L^2(S_R \times (0,h);\mathbb{C}) \oplus L^2(S_R \times (h,h+d);\mathbb{C}^2).$$

Define J_θ to map the absorber scalar to the x component and the plate vector through D_θ into the xy plane. It is a contraction for $0 \le \theta \le 1$, and is continuous in norm. It need not be injective at zero. Set

$$\alpha(\omega) = \operatorname{diag}(\alpha_A(\omega), C_\gamma(\omega)/N), \quad \chi_\theta = J_\theta \alpha J_\theta^*, \quad L_{R,\theta} = \alpha^{-1} - J_\theta^* K_{D_R}(k) J_\theta.$$

The material weight is

$$W = -\operatorname{Im}_H \alpha^{-1} = \alpha^{-*}(\operatorname{Im}_H \alpha)\alpha^{-1} \succ 0. \tag{14}$$

For this fixed stage, compactness of B gives a uniform constant $c_* > 0$ such that $W \succeq c_* I$. The Green radiation identity implies

$$-\Im\langle r, L_{R,\theta} r\rangle = \langle r, Wr\rangle + \Im\langle J_\theta r, K_{D_R} J_\theta r\rangle \ge c_* \|r\|^2 .$$

The coercive sesquilinear form yields invertibility and $\left\|L_{R,\theta}^{-1}\right\| \le c_*^{-1}$ independently of R, ω, θ. This argument does not require J_θ to be an isometry.

The projected incident field is exactly $J_\theta^* f_\pm = \theta g_\pm$, where $g_\pm$ is zero in the absorber and $e_y \mathrm{e}^{\pm ikz}$ in the two-component plate space. Therefore

$$r_{R,\theta}^\pm = L_{R,\theta}^{-1} g_\pm, \quad p_\pm = \theta J_\theta r_{R,\theta}^\pm, \quad X_R = \theta^2 Y_R, \quad A_R^\pm = \theta^2 B_R^\pm, \tag{15}$$

with

$$Y_R = k\Im\left\langle g_\pm, r_{R,\theta}^\pm\right\rangle, \qquad B_R^\pm = k\left\langle r_{R,\theta}^\pm, Wr_{R,\theta}^\pm\right\rangle, \qquad C_R^\pm = Y_R - B_R^\pm.$$

For example, with the physical auxiliary current $q = \alpha J_\theta^* E$, one has $q = \theta r$ and $E^*(\operatorname{Im}_H \chi_\theta)E = q^* W q$. This also proves the absorption identity. The normalized quantities extend continuously to zero coupling through L^{-1}, even though the physical incident response then vanishes.

5 Frequency-uniform finite-aperture localization

Proposition 1. *Freeze positive thicknesses and continuous active matrix laws with* $\operatorname{Im}_H \alpha \succ 0$ *on* B*. On every compact coupling interval in* $[0,1]$*,*

$$\sup_{\omega,\theta} R^{-1} \left\| r^{\pm}_{R,\theta} - r^{\pm}_{\infty,\theta} \right\|_{\mathcal{H}_R} \longrightarrow 0,$$

where the planar current is extended constantly across the aperture. Uniformly on the same parameters,

$$R^{-2}(Y_R, B^{\pm}_R, C^{\pm}_R) \longrightarrow (Y_\infty, B^{\pm}_\infty, C^{\pm}_\infty).$$

The contrast ratios converge uniformly wherever Y_∞ *has a positive lower bound; here it does on each compact parameter set.*

Proof. Let $K_\kappa(k)$ be the transverse Weyl fiber on the fixed height interval, $\beta = (k^2 - |\kappa|^2)^{1/2}$ with outgoing branch. Its kernels contain $G_\kappa(z) = ie^{i\beta|z|}/(2\beta)$ and its first derivatives, together with the longitudinal local term. The explicit bounds from the base paper are uniform for $k \in [k_1, k_2]$, where $k_i = \omega_i / c_{\mathrm{v}}$:

$$\|K_\kappa\| \le C\left(1 + |\kappa| + \frac{k^2 + |\kappa|^2}{|\beta|}\right), \qquad \|K_\kappa - K_0\| \le C_1 |\kappa| \quad (|\kappa| \le k_1/2).$$

Projection by J_θ cannot increase these bounds. The low disk is fixed independently of frequency.

For a fixed smooth window v supported in the unit square and $v_R(x) = v(x/R)$, the Fourier transform of the localization commutator is

$$R^2 \widehat{v}(R\kappa)(J^*_\theta K_\kappa J_\theta - J^*_\theta K_0 J_\theta) r.$$

The low disk has global L^2 bound $C\|r\| \|\nabla v\|_2$ by Plancherel. For the complement, Schwartz decay and an integer $m > 3$ give $R^2 |\widehat{v}(R\kappa)| \le C_{v,m} R^{2-m} |\kappa|^{-m}$. Its fiber bound is uniformly integrable in ω: on $k_1/2 \le \rho \le 2k_2$,

$$\left|k^2 - \rho^2\right|^{-1/2} \le k_1^{-1/2} |\rho - k|^{-1/2},$$

whose radial integral is uniformly finite as the grazing circle moves. For $\rho > 2k_2$, the radial tail is $O(\rho^{2-m})$. Fourier L^1 inversion thus bounds the high-frequency part in transverse L^∞ by $C_{v,m} R^{2-m} \|r\|$, and its aperture L^2 norm by $C_{v,m} R^{3-m} \|r\|$. No square of the grazing singularity is used.

For completeness, the transverse zero fiber is $K_0 p(z) = (ik/2) \int e^{ik|z-z'|} p(z')\, dz'$. It satisfies

$$\Im \langle p, K_0 p \rangle = \frac{k}{4}\left(\left\| \int e^{ikz} p(z)\, dz \right\|^2 + \left\| \int e^{-ikz} p(z)\, dz \right\|^2\right) \ge 0.$$

Thus the same coercivity proof applies to the planar projected operator. The planar inverse has the same c_*^{-1} bound and its currents have uniformly bounded height norm. Applying the inverse to the windowed-current residual gives

$$\sup_{\omega,\theta} \frac{\|r_R - r_\infty\|_{\mathcal{H}_R}}{R} \le C \|1 - v\|_{L^2(S_1)} + \frac{C_v}{R} + O(R^{2-m}).$$

First let $R \to \infty$ for fixed v, then approximate one in $L^2(S_1)$. Density identifies the distributional Fourier construction with the restricted volume field, as in the base proof. Uniform bounds on

k, W, g and currents of norm $O(R)$ give the work and absorption limits by Cauchy–Schwarz; energy balance gives the scattering limit.

Finally, the planar kernels, active matrices and drives are continuous in (ω, θ) in operator or Hilbert norm. The inverse identity gives continuous planar currents. They are nonzero because $g \neq 0$ and L is invertible. Hence $Y_\infty \geq k \langle r, Wr \rangle > 0$ pointwise, and continuity on the compact parameter set supplies a positive minimum. This justifies uniform division. □

6 One center-frequency tuning and strict tensor loss

We now prove the proposed theorem, using a tight bracket so that the floor holds throughout it. Put $q = E_{\min}/E_* < 1$. Choose the oscillator shaping accuracy so that Y_0 is sufficiently close to one that $\min_B Y_0/Y_0(\omega_c) > q$. Choose finite N and then positive damping as above, retaining strict material-cap margins, so that the fixed active stage has

$$m := \min_B Y_\infty(\omega, 0) > 0, \qquad \frac{m}{Y_c} > q, \qquad \inf_B \gamma_\infty(\omega, 0) > 1 - \epsilon/4, \quad Y_c := Y_\infty(\omega_c, 0).$$

Choose two fixed positive coupling-scale constants $s_- < s_+$ with

$$\sqrt{E_{\min}/m} < s_- < \sqrt{E_*/Y_c} < s_+. \tag{16}$$

In particular $s_-^2 m > E_{\min}$, while the center-frequency extinctions at the two bracket endpoints straddle E_*. The bracket for a physical aperture is $\theta \in [s_-/R, s_+/R]$.

By uniform continuity and Proposition 1, for all sufficiently large finite R, this entire bracket lies in $[0, 1]$ and satisfies, with strict margins,

$$X_R(\omega, \theta) > E_{\min} \quad (\omega \in B), \qquad \Gamma_R(\omega, \theta) > 1 - 3\epsilon/4,$$

and

$$X_R(\omega_c, s_-/R) < E_* < X_R(\omega_c, s_+/R).$$

Indeed, uniformly in ω and $s \in [s_-, s_+]$,

$$R^{-2} Y_R(\omega, s/R) \to Y_\infty(\omega, 0), \qquad X_R(\omega, s/R) = s^2 R^{-2} Y_R(\omega, s/R).$$

The second relation is exact. The factor-two bracket from the single-frequency paper would not in general preserve an arbitrarily high floor; (16) supplies the needed replacement.

The intermediate tensor is only semidefinitely lossy. For the selected finite R, add the same isotropic passive oscillator $\nu f_0(z) I$ to each layer, where

$$f_0(z) = \frac{F_0}{\Omega_0^2 - z^2 - i\gamma_0 z}, \qquad F_0, \gamma_0 > 0, \quad \Omega_0 > \omega_2.$$

The full-vector electric inverse before this addition is

$$(I - K_{D_R} J_\theta \alpha J_\theta^*)^{-1} = I + K_{D_R} J_\theta L_{R,\theta}^{-1} J_\theta^*.$$

For fixed R, $K_{D_R}(k)$ is norm-continuous and uniformly bounded on B: its static singular part is independent of k, and its remaining weakly singular kernel varies continuously on the bounded domain. In detail, $g_k - g_0 = ik/(4\pi) - k^2 |x| / (8\pi) + O(|x|^2)$ near zero. The second derivatives of this difference and $k^2 g_k$ are $O(|x|^{-1})$, uniformly on compact k intervals; an integrable Schur bound

gives the asserted norm continuity. If $M_R = \sup_B \|K_{D_R}(k)\|$, the inverse norm is at most $1+M_R/c_*$. Thus

$$\nu \sup_B |f_0|\ M_R(1 + M_R/c_*) < 1$$

gives a uniform Neumann perturbation on the band and bracket. Choosing $\nu > 0$ still smaller preserves all strict floor, contrast and center-endpoint margins. Positive real part of f_0 preserves the lower real cap; the preallocated upper margins preserve the upper caps; its positive imaginary part gives strictly positive tensor loss in every direction.

The final extinction is continuous in θ. A single intermediate-value argument at ω_c supplies a fixed θ in the bracket with $X(\omega_c) = E_*$ exactly. Since the floor and contrast inequalities hold throughout the bracket and band, they hold for this one retuned body. Every selected thickness, aperture, oscillator count and parameter is finite. The order is essential: shape, choose finite weak/thick stage, give positive active damping, choose a tight bracket, enlarge the aperture, then add final isotropic loss and retune. No uniform estimate in all these limits jointly has been invoked.

Passivity gives $\Gamma \leq 1$. At any frequency, equality for a compact body with positive extinction would require one incidence to have zero total scattering and absorption equal to extinction. Its continuous far field would then vanish everywhere, contradicting the forward optical theorem. Therefore $\Gamma(\omega_c) < 1$ for every admissible body, so the worst-case supremum cannot be attained. This completes the analytical construction.

7 Reproducible planar checks and their limits

The accompanying `planar_checks.py` uses NumPy and exact homogeneous slab/cascade formulas, with separate standing-wave quadrature of volume loss. The numerical output is saved in `numerical_results.json`. These tests sample frequency and check algebra; they are not interval certificates or a three-dimensional finite-aperture calculation.

For a moderate damped tensor, independent volume absorption and outgoing-flux absorption agree within 2.9×10^{-16}; opposite-incidence extinction agrees at the displayed precision. A separate lossless matrix-plate check has maximum unitarity residual 9.4×10^{-16} over 31 frequencies. For the moderate lossless plate matrix $\left(\begin{smallmatrix} 1 & 0.4 \\ 0.4 & 1.5 \end{smallmatrix}\right)$, the exact slab plus damped absorber gives the following values over a 21-point grid on $[1, 2]$ at $\theta = 0.003$:

N	$\max \lvert Y - Y_0 \rvert$	$\min \Gamma$
10	0.027083	0.587407
30	0.009758	0.935600
100	0.003172	0.999431
300	0.001259	0.999437

The $\theta \to 0$ limit is essential: increasing N alone at a fixed nonzero coupling need not drive the contrast all the way to one.

A *separate* rational PSD shaping calculation uses eight poles and 501 frequencies on $[1, 2]$, with $\omega_c = 1.5$ and target $Y_0 = 1$. Its ideal normalized coefficient ranges from 0.997495 to 1.001077, with sampled floor-to-center ratio 0.996766. The fixed-axis comparison has ratio 0.75. The compensation cost is large: $\max c \approx 1.2464\times10^{14}$, while $\max |a - 1| \approx 1.04\times10^{-5}$. This shaped example evaluates (13); it is not the moderate matrix used in the finite-stack test, and no finite device dimensions are claimed for it. The analytic argument allows the subsequent scales to be arbitrarily large but finite. These calculations do not provide useful engineering resource bounds.

8 What the draft does and does not settle

The proposed theorem answers the resource-unrestricted fixed-band question for every strict floor $0 < E_{\min} < E_*$. The tensor shaping is within the stated two-layer material class, but goes beyond the fixed-principal-axis rank-one intermediate ansatz used in the original proof.

For that simpler ansatz, a weak plate can give $q_W = t_W/\mathrm{e}^{ikd} \to \mathrm{e}^{i\tau\omega}$ and hence $Y_0 = 4\sin^2(\tau\omega/2)$. Center tuning then guarantees only floor ratios strictly below

$$\rho_0 = \frac{\sin^2(\pi\omega_1/(\omega_1+\omega_2))}{\sin^2(\pi\omega_c/(\omega_1+\omega_2))}.$$

This is a useful restricted-class result, not a bound on the full tensor class.

When $E_{\min} = E_*$, the center must be a global minimum of the exact finite-body extinction spectrum. Uniform approximation to a flat spectrum and a single center tuning do not ensure that condition. The present argument establishes neither existence nor a strict contrast gap in that boundary case. Exactly constant extinction throughout B is a stronger unresolved condition. Fixed-resource optima, quantitative resource rates and an all-isotropic realization also remain outside this construction.

The inherited single-frequency formalization does not verify the new rational density lemma, matrix aperture extension or broadband selection argument. Numerical planar calculations can check algebra and illustrate convergence; they cannot certify the finite three-dimensional Maxwell limit. No novelty or human-peer-review claim is made.

References

[1] D. Wang and J. Zheng, *Unit Supremum of Absorption Asymmetry at Prescribed Extinction for Compact Reciprocal Dielectrics*, manuscript dated 17 October 2026. Source revision `22fed75`, Sections 3, 5–9 and Appendix B. `https://dan-k391.github.io/oq7-papers/prescribed-extinction/`.

[2] L. Zhang, F. Monticone and O. D. Miller, *All electromagnetic scattering bodies are matrix-valued oscillators*, Nature Communications **14** (2023). `https://doi.org/10.1038/s41467-023-43221-2`.